

Authors' Responses

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Authors:	Anonymous Authors

We sincerely thank the Editor and the reviewers for their careful evaluation of our manuscript and for their constructive comments. Their feedback has helped us identify where the contribution, safety claims, experimental evidence, and implementation details require clearer presentation.

We are preparing a major revision of the manuscript. In the revised version, we will:

- 1) clarify the specific contribution of VLM-guided threshold adaptation through comparisons with conventional fixed and adaptive alternatives;
- 2) revise the safety terminology and explicitly delimit the empirical, task-specific scope of the reported safety results;
- 3) strengthen the evaluation of the Bellman critic, teacher policies, real-world experiments, and resource usage; and
- 4) provide the implementation details needed to reproduce the VLM-based threshold tuner and the intervention pipeline.

Detailed responses to each comment are provided below. Reviewer comments are reproduced in black, and our responses are shown in blue. Red bracketed text marks details that will be completed after the corresponding revision or experiment has been finalized. Changes in the revised manuscript will be highlighted in blue.

A Appendix Materials

A.1 RL Teacher Training and Safety Details

Table 1 contrasts the actor’s noisy, deployable signals with the centralized critic’s clean, privileged simulator state. The palm and five fingertips constitute the six hand links. A “twist” is the concatenation of three-dimensional linear and angular velocity. Running mean/variance normalization is applied to both inputs and to value targets.

Table 1: Asymmetric teacher observations.

Signal	Actor	Critic	Definition
Robot joint position q_t	13 (noisy)	13	Actuated arm–hand joint angles (rad)
Robot joint velocity $\dot{q}_t$	13 (noisy)	13	Actuated arm–hand joint velocities (rad/s)
Hand-link position	18 (noisy)	18	Palm and five fingertip positions: 6×3 Cartesian coordinates
Hand-link velocity	18 (noisy)	36	Actor: linear velocity (6×3); critic: spatial twist (6×6)
Palm contact force	–	3	Privileged Cartesian contact-force vector (N)
Measured joint torque	–	19	Privileged torque measurements for 19 instrumented joints (N m)
Object position	3 (noisy)	3	Object Cartesian position
Object orientation	4 (noisy)	4	Object rotation as a unit quaternion
Object velocity	–	6	Privileged object linear and angular velocity (twist)
Goal position	3	3	Cartesian target position for the lift task
Object scale	1	1	Scalar geometry scale supplied by the environment
Object ID	1	1	Scalar identifier for the current object
Previous action a_{t-1}	13	13	Previous 13-dimensional arm–hand command
Total dimension	87	133	

“–” denotes a signal that is unavailable to the actor. Noisy entries are sensor-like estimates; unmarked actor entries are task context or action history. Critic-only entries are used during simulation training and are not available to the deployed policy.

Termination conditions. All conditions are checked at 60 Hz in the environment frame; the table-top height is $z_T = 0.25$ m.

- (a) *Harmful collision:* We define a safety-clearance region above the table and monitor all arm–table contacts. We trigger this termination when the palm or fingertips enter the clearance region, or when any arm link contacts the table.
- (b) *Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.
- (c) *Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90° , at which point the nominal grasp geometry is inverted.
- (d) *Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.
- (e) *Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we use the labeling priority object escape $\succ$ hand too far $\succ$ harmful collision $\succ$ palm flipped $\succ$ timeout. *Requested by Reviewer #22.*

Runs and random seeds. Teacher performance in Main Paper Table II is evaluated over 200 episodes per object, and the real-world results are evaluated over 30 episodes per object. Each of the 40 object-specific teacher policies is trained with five independent random seeds $\{1, 2, 3, 4, 5\}$, with the same evaluation protocol applied to every seed. For each seed, teacher performance is evaluated on all 40 objects. The table below reports the mean $\pm$ standard deviation across the 40 object-level rates for lift success, overall unsafe events, and the four unsafe-event categories separately for each seed. *Requested by Reviewers #14 and #22.*

Table 2: Teacher policy performance across all 40 objects for each random seed (200 evaluation episodes per object and seed). Each entry is the mean $\pm$ standard deviation across the 40 object-level rates. Numerical entries are provisional synthetic placeholders pending verification with the measured evaluation data.

Metric (%)	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5
Lift success	86.7 ± 7.8	87.4 ± 7.5	86.9 ± 8.0	87.8 ± 7.3	87.1 ± 7.7
Unsafe rate avg.	14.2 ± 6.1	13.6 ± 5.8	14.0 ± 6.3	13.2 ± 5.6	13.8 ± 6.0
Harmful collision	3.1 ± 3.5	2.9 ± 3.3	3.2 ± 3.6	2.8 ± 3.2	3.0 ± 3.4
Object escape	9.0 ± 5.2	8.8 ± 5.0	8.9 ± 5.3	8.7 ± 4.9	8.9 ± 5.1
Palm flipped	1.2 ± 2.4	1.1 ± 2.3	1.1 ± 2.3	1.0 ± 2.2	1.1 ± 2.3
Hand too far	0.9 ± 1.9	0.8 ± 1.8	0.8 ± 1.8	0.7 ± 1.7	0.8 ± 1.8

Teacher safety and disagreement handling. Main Paper Table II reports representative teacher unsafe-event rates. Across all 40 teacher policies, unsafe-event rates range from 5–30%. Because an intervention transfers control to an imperfect teacher, the framework’s safety performance is restricted by the teacher’s own performance ceiling. Teacher takeover can reduce unsafe exploration, but it does not provide a formal safety guarantee. The controller does not screen the teacher and student as two symmetric candidates. Instead, the Bellman critic evaluates only the student state–action pair, and the deterministic intervention gate is

$$g_t = \mathbb{I}[\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q], \quad a_t = (1 - g_t)a_t^S + g_t a_t^T.$$

Consequently, disagreement between the two intervention signals is resolved conservatively: either excessive teacher–student action disagreement or a low student success-value prediction is sufficient to trigger teacher takeover. The student action is executed only when both tests accept it. Once triggered, teacher control is retained for a preset minimum dwell interval to avoid rapid switching between policies.

Because the critic is not used to veto the teacher action, the implemented gate has no branch in which both candidate actions are rejected. The teacher action is the terminal action-level fallback whenever $g_t = 1$, even though the teacher is empirically rather than certifiably safe. If the resulting rollout reaches any monitored unsafe condition, the simulator terminates and resets the episode according to the termination rules above; the controller does not synthesize a third recovery action. Non-finite policy outputs are treated as a runtime error before the environment step rather than being accepted by the gate. This design makes the teacher’s performance ceiling an explicit limitation of the intervention mechanism. *Requested by Reviewer #22.*

A.2 VLM Prompt and Inference Details

Visual-input storage and example. Candidate visual inputs are retained during a run in a bounded in-memory buffer of at most 64 samples. At each query, up to eight selected samples are embedded as JPEG data URLs in a self-contained `<run_dir>/vlm_reports/*.html` report together with their metadata, the prompt, model reply, parsed JSON, and API diagnostics. The accompanying `vlm_threshold_advisor.jsonl` strips the base64 payloads and retains only metadata; no separate image files are written.

Figure 1 shows four actual inputs extracted from the first archived query report, one for each object in that rollout.

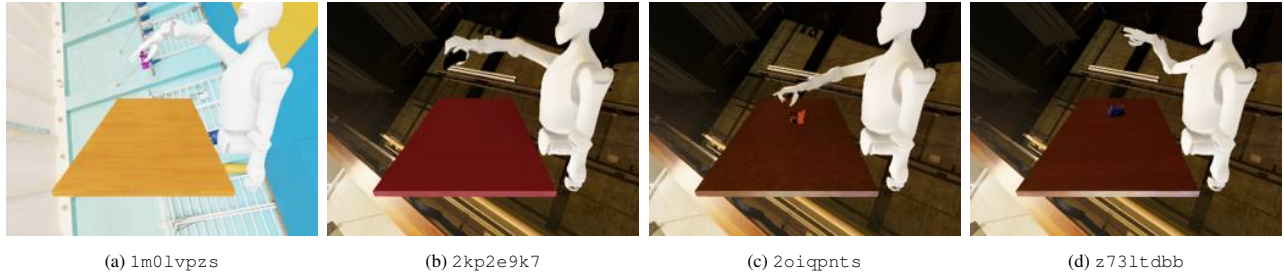


Figure 1: Representative left-camera RGB inputs embedded in `step_000002000_attempt_0001_success.html`. Samples (a)–(c) were selected for high teacher–student disagreement, with ℓ_2 values 9.7506, 17.0997, and 51.9710, respectively; (d) was selected for low predicted success (0.2734). All four records have `unsafe=true`. The transmitted images are 256×192 JPEGs (quality 70). This archived call tagged the images as `detail="low"`; new calls transmit the same image format with `detail="high"`.

Requested by Reviewers #14 and #22.

The following is the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

System message: You tune safety thresholds and return compact JSON only. User message: You are a conservative threshold advisor for DexSafeDagger robot distillation. Do not decide intervention directly. Recommend thresholds for the existing gates. Task: Tune teacher–student disagreement and predicted–success thresholds for dexterous robot distillation. Existing arbitration: - intervene if <code>teacher_student_l2 >= l2_threshold</code> - or if <code>predictor_success <= success_threshold</code> Return only strict JSON with keys: <code>l2_threshold</code>, <code>success_threshold</code>, <code>confidence</code>, <code>reason</code>. Keep changes conservative. Prefer small threshold shifts unless the recent window shows clear over- or under-intervention. Base thresholds and clamps: - <code>base_l2_threshold={base_l2_threshold}</code> - <code>allowed_l2_range=[{l2_min}, {l2_max}]</code> - <code>base_success_threshold={base_success_threshold}</code> - <code>allowed_success_range=[{success_min}, {success_max}]</code> Current advisor thresholds: - <code>current_l2_threshold={current_l2_threshold}</code> - <code>current_success_threshold={current_success_threshold}</code>	Recent rollout statistics: <pre>{ "frame": {frame}, "intervention_rate": {beta}, "step": {step}, "sample_count": {N}, "l2_threshold": {tau_delta}, "success_threshold": {tau_Q}, "unsafe_episode_rate": {r_unsafe}, "l2": {"min": {v}, "max": {v}, "mean": {v}, "p10": {v}, "p50": {v}, "p90": {v}}, "success": {"min": {v}, "max": {v}, "mean": {v}, "p10": {v}, "p50": {v}, "p90": {v}}, "unsafe_reason_prop": { "hand_too_far": {v}, "harmful_collision": {v}, "object_out_of_bound": {v}, "palm_flipped": {v}}, "visual_buffer_size": {B}, "visual_samples_attached": {M}, "visual_samples": [{ "env_id": {v}, "frame": {v}, "image_attached": true, "image_key": {v}, "source": {high_l2 low_success unsafe_triggered}, "l2_threshold": {v}, "success_threshold": {v}, "teacher_student_l2": {v}, "predictor_success": {v}, "object_id": {v}, "object_name": {v}, "object_env_count": {v}, "step": {v}, "unsafe": {true false}}, ...] }</pre> For each selected visual sample $i=1, \dots, M$: Visual sample {i} metadata: {the corresponding record above, excluding "image_attached"} <IMAGE {i}: JPEG data URL, <code>detail="high"</code>>
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The earlier abbreviated prompt omitted the distribution summaries, threshold clamps, unsafe-reason breakdown, visual-buffer context, and per-image metadata; it also used the nonexistent output key `disagreement_threshold` and omitted `confidence`. The runtime prompt uses `l2_threshold`. For reproducibility, the template above preserves the literal inclusive comparison symbols sent to the VLM. The executed controller uses the strict tests in the intervention equation above; the distinction affects only exact equality at a threshold and does not change the VLM’s role as a threshold advisor.

Requested by Reviewers #14 and #22.

Model and input configuration. We used the OpenAI GPT-5.5 multimodal model through the Chat Completions endpoint (`/v1/chat/completions`, `model="gpt-5.5"`). Each request set `stream=false` and `max_completion_tokens=1024`; no temperature or reasoning-effort parameter was sent, so the provider defaults applied. The model received the textual rollout statistics above and up to eight RGB images, encoded as

downsampled JPEG data URLs with high visual detail. We issued the first VLM query after 2,000 online distillation steps and subsequently queried it every 10,000 steps, resulting in 10 calls over the 100,000-step run. The archived API reports, which were generated with the previous low-detail image setting, record 30,641 prompt tokens and 4,841 completion tokens, or 35,482 tokens in total. *Requested by Reviewers #14 and #22.*

Output processing and threshold updates. The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "l2_threshold": <number>, "success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The implementation first attempts to parse the complete assistant response as JSON. If that fails, it parses the substring from the first opening brace to the last closing brace, which permits recovery from a surrounding Markdown code fence or brief prefatory text. Only a JSON object is accepted. The two threshold fields are converted to floating-point values independently; a missing or non-convertible field leaves that threshold at its current value, and unrecognized fields are ignored. The `confidence` and `reason` fields are recorded for diagnostics but neither changes nor vetoes the controller update.

The returned thresholds are interpreted as absolute candidates, not direct intervention decisions. For each $x \in \{\delta, Q\}$, the candidate is clipped and exponentially smoothed as

$$\bar{\tau}_x^{(k)} = \text{clip}\left(\hat{\tau}_x^{(k)}, c_x^{\min} \tau_x^{(0)}, c_x^{\max} \tau_x^{(0)}\right), \quad \tau_x^{(k+1)} = (1 - \alpha) \tau_x^{(k)} + \alpha \bar{\tau}_x^{(k)}.$$

We used $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. In the archived run, $\tau_\delta^{(0)} = 3.0$, yielding the allowed range $[1.5, 4.5]$. Warm-start calibration set $\tau_Q^{(0)} = 0.7081624$, yielding the implemented range $[0.3540812, 1.0622436]$. For example, the first recommendation $\hat{\tau}_\delta = 3.2$ produced the applied value $0.9(3.0) + 0.1(3.2) = 3.02$.

In `active` mode, the filtered disagreement threshold becomes the controller’s τ_δ , and the filtered success threshold is assigned to the Bellman critic’s intervention threshold. They affect the next control step through the strict gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q therefore increases teacher intervention. In `shadow` mode, the same outputs are logged without changing either threshold. The first query was made at online distillation step 2,000 and later queries every 10,000 steps, at steps 2,000, 12,000, ..., 92,000.

The deterministic fallback is to retain the last applied thresholds. This occurs when the API key is unavailable, the request times out or fails, the API response is malformed, or no JSON object can be parsed. If only one threshold field is missing or non-convertible, that coordinate is retained while a valid coordinate may still be clipped and updated. A failed attempt is logged and is not retried immediately; the next attempt occurs at the next scheduled update. Finally, the VLM and this adaptation mechanism are used only during simulation-based online distillation. There is no real-world student training, and during physical deployment the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. *Requested by Reviewers #14 and #22.*

A.3 Additional Experimental Results

Bellman critic evaluation. [State how safety risk and task success are separated, or justify the retained target. Report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate for unsafe episodes at the deployed threshold.] Requested by Reviewers #14 and #22.

Training and resource comparison. All ablations use the same training and computational resource budget and differ only in the intervention mechanism. Requested by Reviewer #22.

Twenty-two-object inventory. The complete evaluation set comprises the following 22 objects (the six objects reported in the main paper are marked with an asterisk):

1	WindCart*	12	Ogre
2	Boot*	13	Cheburashka
3	ToolBox*	14	Lucy statue
4	Cow*	15	Beethoven bust
5	Plane*	16	Igea bust
6	Shoe*	17	Dog pair
7	Horse	18	Spot dog
8	Teapot	19	Suzanne monkey head
9	Sphinx	20	Blub character
10	Dragon	21	Happy character
11	Homer bust	22	Dinosaur

Complete real-world evaluation. We will report results for the remaining 16 real-world test objects that are not shown in the manuscript’s six-object table, using the same success and safety breakdown. The two eight-object blocks below separate task success from the observed unsafe failure modes. The entries below are provisional synthetic values for layout checking and will be replaced with the measured results. Requested by Reviewers #14 and #22.

Table 3: Provisional per-object results for the remaining 16 objects (30 episodes per object). Synthetic values are shown for layout checking only; **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Cheburashka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	13.3	23.3	20.0	26.7	16.7	10.0	23.3
Harmful collision	3.3	0.0	10.0	6.7	13.3	3.3	3.3	6.7
Object escape	13.3	13.3	13.3	13.3	10.0	13.3	6.7	13.3
Palm flipped	0.0	0.0	0.0	0.0	3.3	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	Dog Pair	Spot	Suzanne	Blub	Happy	Dinosaur
Lift Success	86.7	73.3	70.0	63.3	73.3	60.0	66.7	56.7
Unsafe Rate Avg	20.0	30.0	33.3	40.0	30.0	43.3	36.7	46.7
Harmful collision	6.7	13.3	16.7	20.0	13.3	23.3	16.7	26.7
Object escape	13.3	13.3	13.3	16.7	13.3	16.7	16.7	16.7
Palm flipped	0.0	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

[Replace all provisional synthetic values with the measured results for the remaining 16 objects. Report 95% confidence intervals and statistical comparisons, and provide aggregate results separately for seen and unseen objects across the full 22-object evaluation.] Requested by Reviewers #14 and #22.

Physical baseline comparison. [Report the feasible matched real-robot baseline using the same objects, initial-condition protocol, and number of trials, including lift success, unsafe-event incidence, uncertainty, and the statistical comparison to VLM-DexSafeDagger.] *Requested by Reviewer #14.*

B Response to the Associate Editor

We thank the Associate Editor for summarizing the central issues raised during review. We address each issue below and provide detailed point-by-point responses to the reviewers in the following sections.

Comment 1: The experiments do not sufficiently separate the contribution of the VLM from the effect of adaptive threshold tuning. Additional comparisons and ablations are needed to clarify whether the VLM itself provides a benefit.

[Response:] We agree that the comparison between a fixed intervention policy and VLM-guided adaptation does not, by itself, isolate the source of the improvement. In the revision, we will add controlled comparisons covering fixed threshold selection, conventional adaptive threshold rules, a learned threshold controller, statistics-only language-model input, and statistics-plus-image VLM input. All methods will use the same teacher, student, critic, training budget, and evaluation protocol. We will report both task performance and intervention cost so that safety improvements cannot be attributed only to increased teacher control. **[Insert the finalized methods, principal quantitative result, and revised table/figure locations.]**

Comment 2: The framework provides empirical intervention rather than a formal safety guarantee, and unsafe episodes remain in the real-world experiments. The scope of the “safety-critical” claim should therefore be clarified and better supported.

[Response:] We agree. The revised manuscript will explicitly state that the framework does not guarantee constraint satisfaction and that safety violations remain possible. We will replace overly strong uses of “safety-critical” with terms such as “safety-aware” or “empirically risk-reducing,” and we will limit our conclusions to the task-specific unsafe events measured in the experiments. **[List the affected sections and the final revised title, if applicable.]**

Comment 3: The Bellman success-value critic, VLM reproducibility, number of real-world trials, baseline coverage, and reporting for all tested objects need to be strengthened.

[Response:] We agree and will expand both the evaluation and the implementation description. The revision will include critic error and calibration analyses, the complete VLM prompt and output-processing procedure, the number of random seeds, matched baseline comparisons, uncertainty estimates, and aggregate as well as per-object real-world results. **[Insert the completed analyses and their manuscript or supplementary-material locations.]**

C Response to Reviewer #14

We sincerely thank the reviewer for the careful assessment of our work and for the constructive suggestions. Below are our point-by-point responses.

Comment 1: The current comparison between DexSafeDagger and VLM-DexSafeDagger compares fixed thresholds with VLM-adapted thresholds and therefore does not show whether the improvement comes from threshold adaptation, language reasoning, or visual information. Please compare cases with and without an LLM/VLM and with and without statistics and/or images.

[Response:] We agree. We will add controlled ablations that separate adaptation, language reasoning, and visual input. In particular, the revision will compare a fixed operating point, a non-language adaptive controller, an LLM receiving rollout statistics only, a VLM receiving rollout statistics and selected images, and a visual-control condition in which the image information is removed or mismatched. These variants will be evaluated under matched training and intervention budgets. **[Insert final ablation definitions, results with uncertainty, and table/figure references.]**

Comment 2: The related work on adaptive intervention should be expanded. In particular, AIM, EnsembleDagger, and HG-Dagger are relevant. Since adaptive intervention can be achieved without a VLM, please clarify how VLM-DexSafeDagger differs from these methods and whether the VLM provides an advantage over simpler adaptive approaches.

[Response:] We agree and will expand the related-work discussion to include AIM, EnsembleDagger, and HG-Dagger. We will distinguish uncertainty-based gating, human-gated intervention, learned intervention proxies, and our setting of autonomous multi-object visuomotor distillation with a two-channel disagreement/value gate. We will also ground the claimed benefit of the VLM in the new adaptive-baseline experiments rather than in a conceptual distinction alone. **[Insert citations and the revised section location.]**

Comment 3: Please provide the VLM prompt and explain how images are selected and how the VLM outputs are used. The Bellman success-value critic should also be evaluated, especially for errors on unsafe cases. Please clarify the number of runs or random seeds and the scope of the term “safety-critical.”

[Response:] We agree. The supplementary material will provide the exact prompt, model identifier, inference settings, image-selection rule, output format, parser, threshold clipping and smoothing, query schedule, retry behavior, and deterministic fallback. We will also report critic discrimination and calibration metrics, with particular attention to false negatives on unsafe episodes, and will state the number of independent runs used for every result. Finally, we will revise the safety terminology as described in our response to the Associate Editor. **[Insert model version, number of seeds, critic metrics, and section/appendix references.]**

Comment 4: Only VLM-DexSafeDagger is evaluated on the real robot. This does not sufficiently demonstrate the proposed concept; a physical baseline comparison is needed.

[Response:] We agree that a matched physical comparison would provide stronger evidence. We will evaluate the distilled policy against **[specify the feasible real-world baseline or baselines]** using the same object set, initial-condition protocol, and number of trials. We will report lift success, unsafe-event incidence, and uncertainty for each method. We will also clarify that the VLM is used during simulation distillation and is not queried by the deployed student policy. **[Insert final results and table reference.]**

Comment 5: The paper states that 22 objects were evaluated, while the current table shows only six representative objects. Please provide overall results for all 22 objects, for example separately for seen and unseen objects.

[Response:] We agree. The revision will report aggregate results for all 16 seen and 6 unseen objects and will provide per-object results for the complete 22-object evaluation in the supplementary material. Each aggregate will include the number of trials and a 95% confidence interval. **[Insert the final seen/unseen results and table references.]**

D Response to Reviewer #22

We sincerely thank the reviewer for the detailed and constructive comments. Below are our point-by-point responses.

Comment 1: The term “safety-critical” is too strong because the framework supplies empirical intervention rather than formal constraint satisfaction. State explicitly that safety violations remain possible.

[Response:] We agree. The revised manuscript will state explicitly that the method provides empirical risk reduction under task-specific event definitions, not a formal safety guarantee, and that nonzero violation probability remains in both simulation and physical deployment. We will revise the title, abstract, main text, and conclusion to remove or qualify claims that exceed this scope. **[Insert the final wording and section locations.]**

Comment 2: Demonstrate that the VLM is necessary by comparing against fixed thresholds, rule-based adaptation, Bayesian optimization, curriculum schedules, grid search, and a small learned threshold network.

[Response:] We agree that stronger conventional baselines are required. We will compare validation-selected fixed thresholds (using grid search and Bayesian optimization) with adaptive curriculum, rule-based feedback, learned-network, and VLM-guided controllers. The fixed methods will select one operating point, whereas the adaptive methods will update the operating point during training. All comparisons will use matched data and compute budgets where applicable and will report task success, unsafe-event rate, intervention rate, teacher queries, and training cost. **[Insert the final protocol, quantitative results, and table/figure references.]**

Comment 3: The Bellman critic predicts combined “safe-lift-success,” thereby conflating task success with safety. Separate safety risk and task value, and report calibration, false-negative rates, and reliability diagrams.

[Response:] We agree that the current joint target does not distinguish an unsafe outcome from a safe episode that fails to complete the lift. In the revision, we will **[state whether the critic is redesigned with separate safety and task heads, or justify and retain the joint target]**. We will report AUROC, AUPRC, Brier score, expected calibration error, reliability diagrams, and the false-negative rate at the deployed threshold on held-out trajectories. **[Describe exactly how the revised critic enters the intervention gate and insert the results.]**

Comment 4: Teacher intervention is assumed safe, yet teacher policies may collide or fail. Quantify teacher unsafe rates and define how the framework responds when teacher and critic disagree.

[Response:] We agree that teacher takeover is not itself a safety guarantee. We will report success and unsafe-event rates for the teacher policies over the complete evaluation set rather than only representative objects. We will also specify how the controller handles disagreement between the critic’s assessment and the teacher action, including the fallback used when neither candidate action satisfies the acceptance rule. **[Insert teacher statistics, the fallback rule, and section/table references.]**

Comment 5: Provide exact VLM prompts, model version, decoding settings, image-selection procedure, output parsing, update frequency, latency, cost, and failure handling. Reproducibility may otherwise depend on an external proprietary service.

[Response:] We agree. The revised supplementary material will include all of the requested information, together with the permitted threshold range, smoothing rule, invalid-output handling, and deterministic fallback. We will archive the raw VLM outputs and threshold trajectories where allowed so that the adaptation process can be audited independently of the external service. In the reported run, all ten scheduled VLM calls completed successfully; we encountered no request timeout, API failure, or malformed VLM output. The implementation details and recorded token usage are provided in Appendix A.2.

Comment 6: Real-world unseen-object performance is weak: lift success falls to 60–66.7%, while unsafe episodes reach 40%. Claims of sufficient deployment robustness should be moderated.

[Response:] We agree. We will remove the claim that the current results establish sufficient robustness for deployment. Instead, we will present the physical experiments as an initial transfer evaluation and identify the unseen-object results as evidence of a remaining sim-to-real and contact-generalization gap. **[Insert the revised discussion and limitation-section locations.]**

Comment 7: Only fifteen real-world episodes per reported object produce coarse estimates. Increase repetitions and report confidence intervals and statistical comparisons across all 22 objects.

[Response:] We agree that the uncertainty of the current estimates must be made explicit. We will **[state the feasible number of additional trials]** and report per-object and aggregate seen/unseen results with 95% confidence intervals. Matched method comparisons will include effect sizes and appropriate statistical tests or bootstrap intervals. **[Insert the final trial count, analysis method, and results.]**

Comment 8: Safety-event thresholds for harmful collision, object escape, palm flipping, and workspace violation require physical justification. Include contact-force measurements and severity-weighted safety metrics.

[Response:] We agree that each event definition requires a physical and sensing rationale. The revision will list the numerical threshold, sensor or simulator signal, units, and justification for every event. In addition to binary episode incidence, we will report **[the available force, impulse, motor-current, penetration, or other severity proxy]**. If calibrated force measurements are not available, we will label the reported signal as a proxy and state this limitation explicitly. **[Define the severity-weighted metric and insert the section/table references.]**

Comment 9: Clarify whether VLM adaptation occurs only during simulation distillation or during physical deployment. If absent at deployment, the role of scene-level VLM reasoning should not be overstated.

[Response:] The VLM is queried only during simulation-based online distillation. During physical deployment, the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. We will state this distinction in the abstract, method, experimental setup, and limitations, and will describe the VLM as a training-time adaptation mechanism rather than a deployment-time safety monitor. **[Insert the final section locations.]**

Comment 10: Compare training samples, teacher queries, interventions, computation, and final success across Vanilla DAgger, SafeDAgger, critic-based intervention, and VLM-guided intervention using multiple random seeds.

[Response:] We agree. We will add a unified comparison using the same training horizon and paired random seeds. For each method, we will report environment steps, student updates, teacher labels, teacher-controlled steps, intervention rate, compute time, VLM calls where applicable, final lift success, final unsafe rate, and unsafe-rate area under the training curve. **[Insert the number of seeds, quantitative results, and table/figure references.]**

We thank the Editor and the reviewers again for their thoughtful feedback. The requested revisions will help us present the method’s contribution more clearly, support its empirical claims more rigorously, and state its limitations more precisely.